Apala Pramanik

Lincoln, Nebraska , USA

apramanik2@huskers.unl.edu | https://apalapramanik.github.io/ | github.com/apalapramanik | linkedin.com/in/apala-pramanik

Personal Profile

A highly motivated PhD student at the School of Computing, University of Nebraska Lincoln. I am focused on millimeter wave communication improvement strategies using machine learning, with a strong foundation in Robotic Cyber-Physical Systems and formal verification methods. *I'm interested in full-time internship opportunities to start in the summer of 2026 in the field of machine learning in wireless communications with applications in V2I/V2X.*

Education

University of Nebraska-Lincoln- PhD — *Nebraska , USA*

Graduate Research Assistant - School of Computing — *June 2024 - Current*

Advised by: Dr. Mehmet Can Vuran, Dale M. Jensen Chair Professor

University of Nebraska-Lincoln- Master's — *Nebraska , USA*

Graduate Research Assistant - School of Computing — *Aug 2021 - May 2024*

Advised by: Dr.Dung Hoang Tran, Assistant Professor

Guru Gobind Singh Indraprastha University- Bachelor's — *Delhi, India*

Electronics and Communication — *Aug 2017 - May 2021*

Thesis: Deep learning techniques for Magnetic Resonance Image (MRI) denoising and reconstruction from undersampled MRI scans.

Research Experience

University of Nebraska–Lincoln — *Nebraska, USA*

Doctor of Philosophy, Computer Science — *Aug 2024 – Current*

- **Camera-Primed Double-Directional Beam Management (IEEE SECON 2026, accepted):** Developed a vision-assisted mmWave beamforming framework enabling real-time TX/RX beam selection using camera-derived spatial cues and adaptive correction under hardware constraints.
- **Reinforcement Learning for mmWave Power Control:** Designed a hardware-in-the-loop RL framework to optimize transmit power and RF gain parameters under SNR constraints using SDR-based testbeds and real-time feedback.
- Utilized ray-tracing tools (**Wireless InSite**, **Sionna RT**) to model propagation environments and validate experimental observations.
- **Technical Skills:** GNU Radio, USRP, Python, C++, Ubuntu Linux, Git

University Of Nebraska-Lincoln — *Nebraska, USA*

Master's, Computer Science — *Aug 2021 - December 2024*

- Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems
- Experience with **Neural Network Verification (NNV)** software for formal analysis of deep learning models.
- Knowledge of **Signal Temporal Logic (STL)** for specifying and verifying temporal properties in cyber-physical systems.
- **Technical Skills:** ROS, Gazebo, RVIZ, HTML, CSS, Python, C++, Ubuntu Linux, Git.

Publications

Conference Proceedings

Look Once, Beam Twice: Camera-Primed Real-Time Double-Directional mmWave Beam Management for Vehicular Connectivity
Avhishek Biswas, Apala Pramanik, Eylem Ekici, Mehmet Can Vuran
Accepted at 2026 IEEE International Conference on Sensing, Communication, and Networking (SECON)(to appear)

Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems
Apala Pramanik, Sung Woo Choi, Yuntao Li, Luan Viet Nguyen, Kyungki Kim, Hoang-Dung Tran
2024 22nd ACM-IEEE International Symposium on Formal Methods and Models for System Design (MEMOCODE), 2024

ASTITVA: assistive special tools and technologies for inclusion of visually challenged
Apala Pramanik, Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Rohan Tripathi
2021 international conference on computing, communication, and intelligent systems (ICCCIS), 2021

START: Smart Stick based on TLC Algorithm in IoT Network for Visually Challenged Persons
Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Apala Pramanik
2020 Fourth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)(I-SMAC), 2020

Workshops

Vision-based Runtime Monitoring for Human-Construction Robot Systems
IROS 2023 Workshop : Formal methods techniques in robotics systems: Design and control
Apala Pramanik, Kyungki Kim, Dung Hoang Tran

University Projects

mmWave Propagation Engineering and Bistatic Radar Characterization *Nebraska, USA*
University of Nebraska–Lincoln *Spring 2025*

- Investigated mmWave propagation mechanisms by leveraging near-critical incidence on dielectric surfaces to induce lateral wave components for extended coverage along walls.
- Designed controlled experiments to characterize angle-dependent reflection and surface-guided propagation behavior in indoor environments.
- Developed a bistatic radar measurement framework using liquid metal (EGaIn) and metallic reflectors to study material-dependent scattering and signal enhancement effects.
- **Technical Skills:** mmWave Propagation, Bistatic Radar, RF Measurement, USRP/SDR, Signal Analysis, Experimental Design
- **Soft Skills:** Research Methodology, Problem Formulation, Data Interpretation, Technical Communication

Multi-Modal Traffic Pattern Detection Using Radio and Camera Sensing *Nebraska, USA*
University of Nebraska–Lincoln *Fall 2024*

- Built a real-time multi-modal traffic classification system using synchronized USRP RF signals and camera images for three-class detection.
- Implemented real- and complex-valued CNN models, achieving $>$80% accuracy with $<$100 ms inference latency.
- Integrated precise timestamp synchronization ($\pm$10 ms) and a watchdog mechanism for reliable, continuous operation.
- **Technical Skills:** Python, PyTorch/TensorFlow, USRP, Signal Processing, Deep Learning (CNN, CVNN), Multi-Modal Data Fusion
- **Soft Skills:** System Design, Cross-Disciplinary Collaboration, Technical Presentation, Analytical Problem Solving

mmWave 3D Ray Tracing Framework *Nebraska, USA*
University of Nebraska–Lincoln *Fall 2024*

- Built a MATLAB-based 3D ray tracing simulator for mmWave indoor propagation with multi-bounce reflections and log-normal shadowing.
- Implemented log-distance path loss modeling and reflection loss computation at 24 GHz.
- Developed modular environment modeling and power-gradient visualization with automated SVG/EPS export and Python-based GIF generation.
- **Technical Skills:** MATLAB, Python, Ray Tracing, Wireless Channel Modeling

Deep Learning and NLP Projects *Nebraska, USA*
University of Nebraska–Lincoln *Aug 2021 – July 2024*

- Developed a Spanish news classification system using GRU, LSTM, and BETO models, achieving 91.80% accuracy with BETO and tokenizer optimization.
- Applied data augmentation techniques (synonym and character replacement) to expand the dataset, improving training accuracy to 99%.
- Implemented image classification models on Fashion-MNIST and CIFAR-100 and performed sentiment analysis using sequential deep learning architectures.
- **Technical Skills:** Python, Pandas, Matplotlib, Seaborn, TensorFlow, Deep Learning (CNNs, RNNs, Transformers)
- **Soft Skills:** Analytical Thinking, Technical Writing, Teamwork, Presentation Skills

Review

IEEE Transactions on Wireless Communications 2026
IEEE Transactions on Wireless Communications 2025

Involvement

Undergrad Research Fair Presented research projects to undergraduate computer science major students
K-12 Outreach Fair Demonstrated research activities to K–12 students
Mentorship Mentored undergraduate researchers on experimental design, data analysis, and research communication

Skills

Programming **Python**: Pandas, PyTorch, Tensorflow, NumPy, Scikit-learn, etc.
ROS: rospy, roscpp
Softwares:GNURadio, Wireless Insite, Matlab, Gazebo, Rviz, Arduino, AVR Studio, Proteus Professional, KEIL, Point Cloud Library
Miscellaneous Linux, Shell (Bash/Zsh), LaTeX(Overleaf/R Markdown), Microsoft Office, Git.
Soft Skills Mentoring, Time Management, Teamwork, Problem-solving, Documentation, Engaging Presentation.

Interests

Paper Review I enjoy critically reviewing research papers to evaluate technical rigor, clarity of methodology, and the strength of scientific contributions.
Technical Writing I enjoy writing research papers to convey my research work in easy and understandable language.
Dancing I have a senior diploma degree in Bharatnatyam (Indian Classical Dance) and have training of more than 10 years.
Fitness I am passionate about weight training and maintaining a healthy and active lifestyle.

References available upon request.